Cheat sheet

Machine Learning

Algorithms, evaluation, workflow and interview essentials

Problem types

Regression	Predict a number (price, demand)
Classification	Predict a category (spam, churn)
Clustering	Group unlabeled data (segments)
Ranking / recommendation	Order items for a user
Anomaly detection	Find rare unusual events (fraud)

Algorithms

Linear regression	Minimise squared error; assumes linearity
Logistic regression	Sigmoid of linear score; log-loss
k-NN	Vote of nearest neighbours; needs scaling
Naive Bayes	Bayes rule with independence assumption; fast text baseline
SVM	Max-margin boundary; kernels for non-linear
Decision tree	Recursive splits by Gini or entropy
Random forest	Bagging + random features
Gradient boosting	Sequential trees on residuals (XGBoost, LightGBM, CatBoost)
K-Means	Minimise within-cluster distance; choose K via elbow / silhouette
DBSCAN	Density clusters, finds noise
PCA / t-SNE / UMAP	Dimensionality reduction / visualisation

Evaluation

Splits	Train / validation / test; stratify for imbalance; time-based split for time series
Cross-validation	k-fold average for robust estimates
Classification	Accuracy, precision, recall, F1, ROC-AUC, PR-AUC, confusion matrix
Regression	MAE, MSE, RMSE, R^2, MAPE
Imbalance	Class weights, resampling (SMOTE), threshold tuning, PR-AUC
Calibration	Predicted probabilities match real frequencies

Bias, variance, regularisation

Underfit	High bias → more features, complex model
Overfit	High variance → more data, regularise, simplify
L1 (Lasso)	Sparse weights, feature selection
L2 (Ridge)	Small weights, stability
Early stopping	Stop when validation stops improving

Workflow

1 Define problem — 2 Data — 3 EDA — 4 Features — 5 Baseline —
6 Model — 7 Tune — 8 Evaluate — 9 Deploy — 10 Monitor drift

Feature scaling	StandardScaler (z-score), MinMaxScaler
Encoding	One-hot, ordinal, target encoding (beware leakage)
Leakage	Fit preprocessing on training data only; time-aware splits
Drift	Data drift (inputs change), concept drift (relationship changes)